計算機組織

final project

CPU Design

指導教授:蔡宗漢

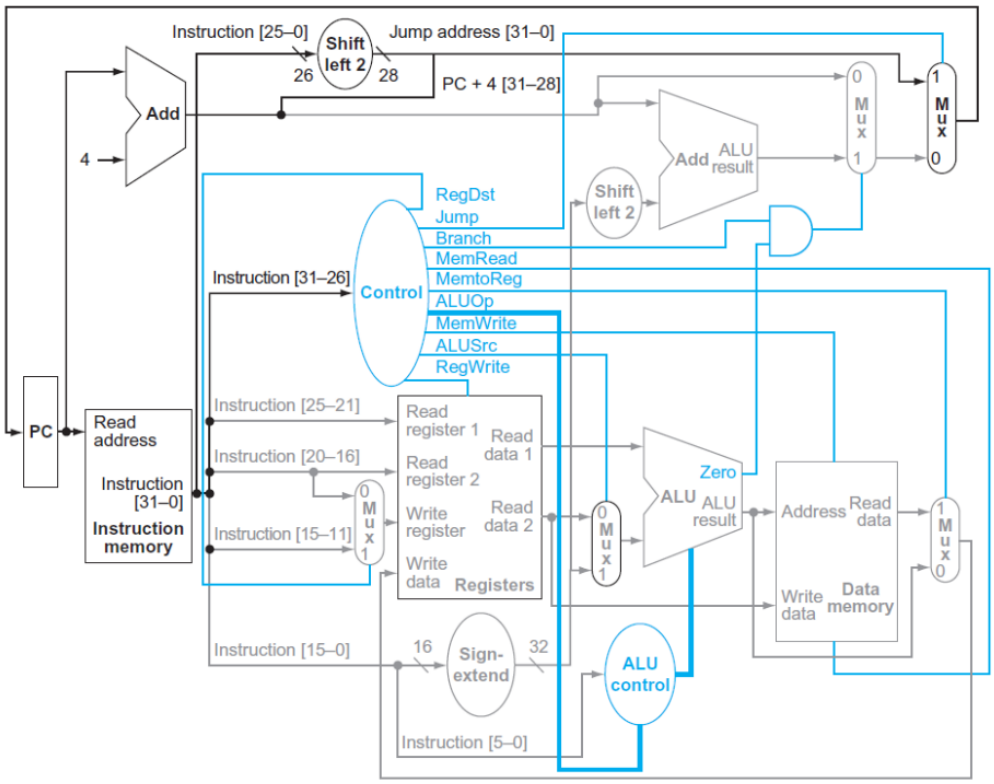
電機 3A: Fira

電機 3B: Tavi

電機 3B: Leto

- Single cycle implementation

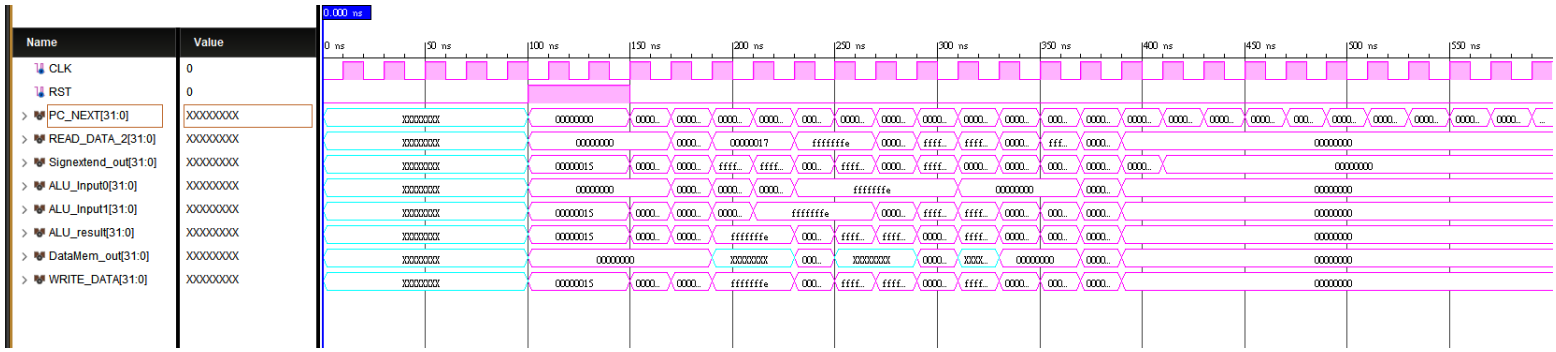
1. Block Diagram



2. 架構說明

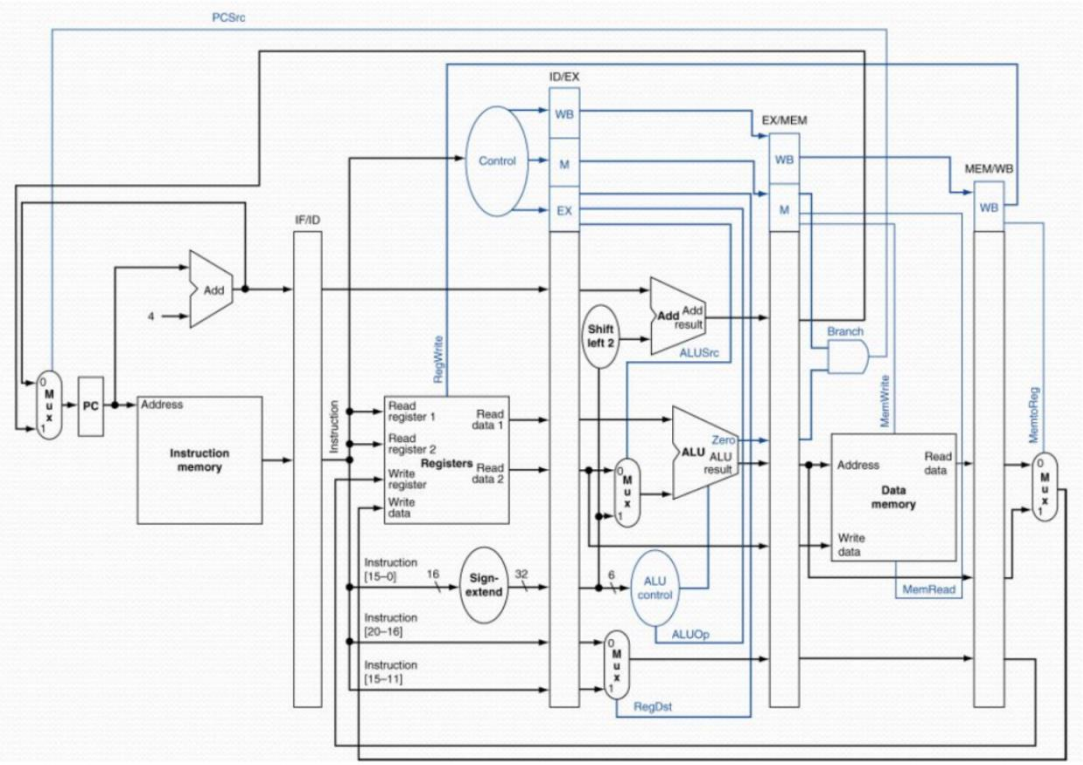
We only need to let the original cycle to wait until IF, ID, EX, MEM, WB are all done before giving the next instruction.

3. 波形圖



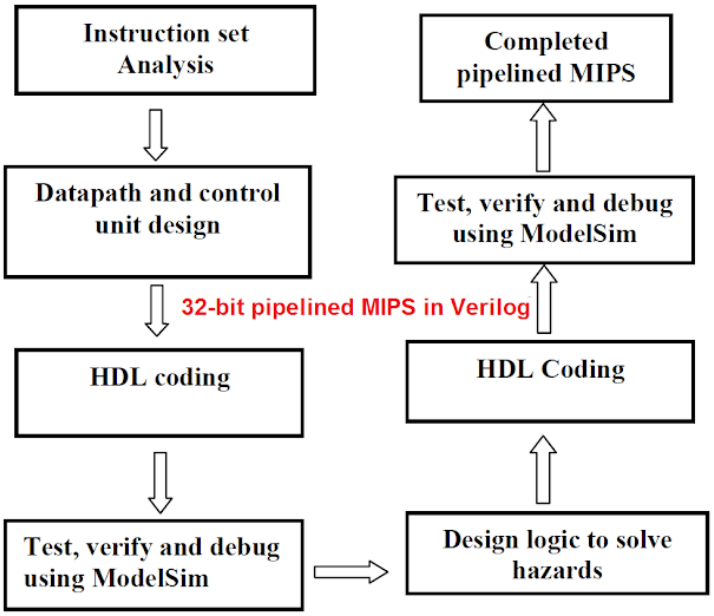
- Pipeline implementation

1. Block Diagram

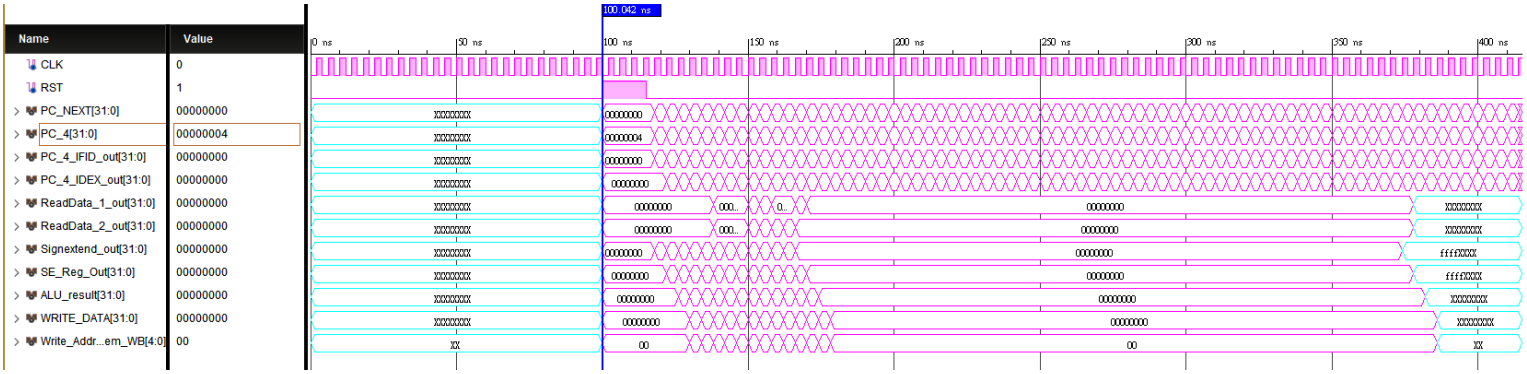


2. 架構說明

The serious drawback is that the clock cycle is determined by the longest possible path in the Processor. Thus, the pipelined MIPS came out to solve those problems by exploiting most functional unit in one clock cycle and improving the performance by increasing the instruction throughput. However, the pipelined MIPS also faces challenges such as control and data hazards.



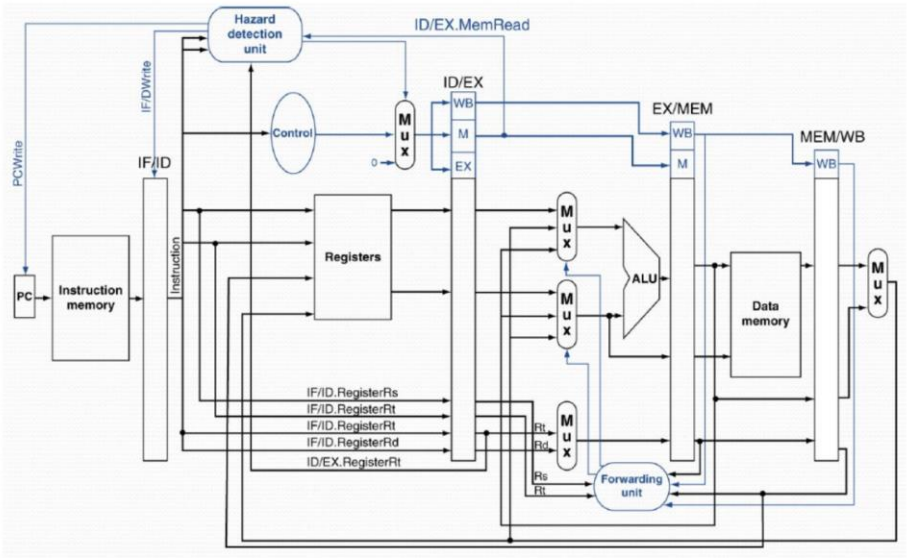
3. 波形圖

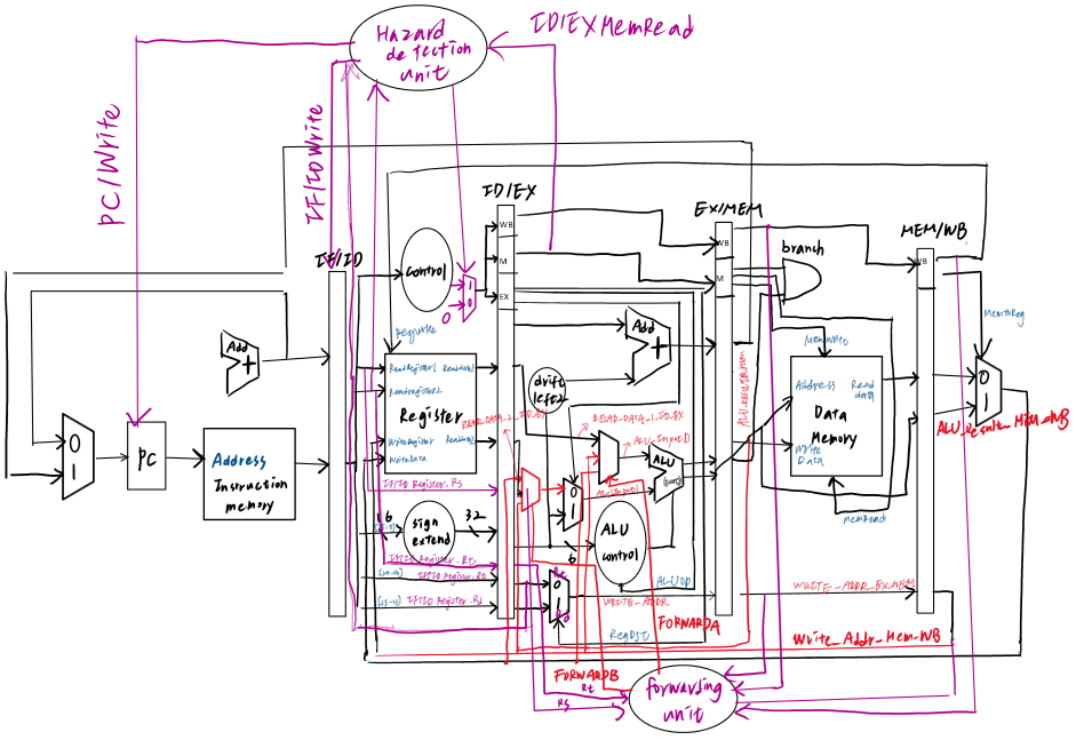


此波形圖因為變化較密，值未直接顯示，實際模擬時游標移動會顯示。

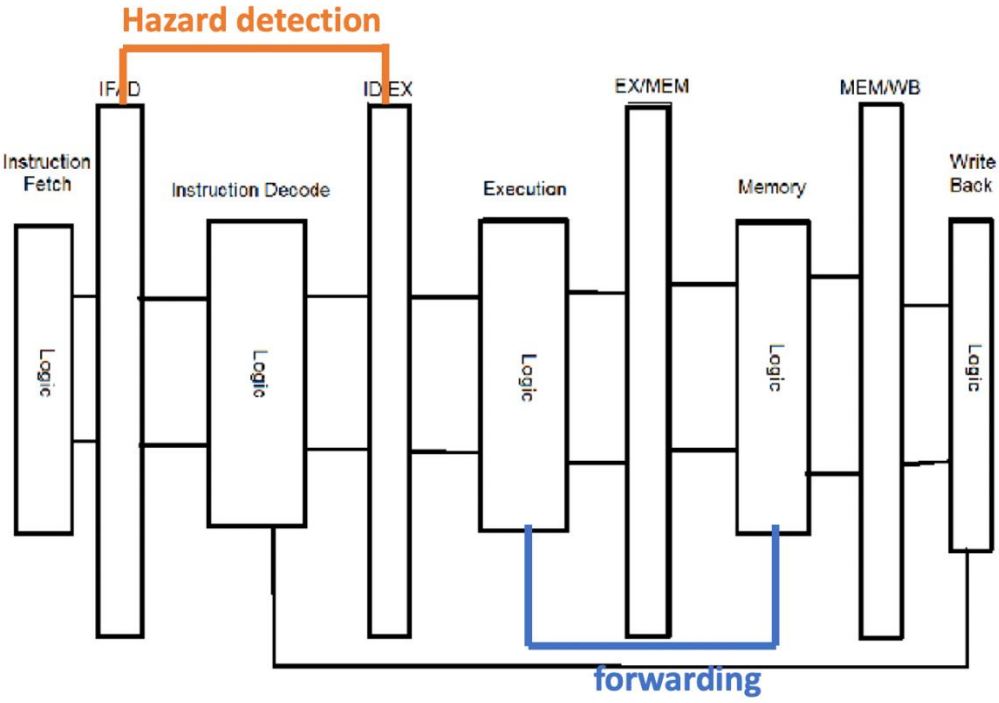
- Hazard detection

1. Block Diagram




Hazard detection unit
IDIEX MemRead
PC Write
IF/ID Write
IF/ID
ID/EX
EX/MEM
MEM/WB
branch
control
Add
PC
Address
Instruction memory
Register
sign extend
ALU
ALU control
Data Memory
FORWARDA
FORWARDB
forwarding unit
Write_Addr_Mem_WB

2. 架構說明

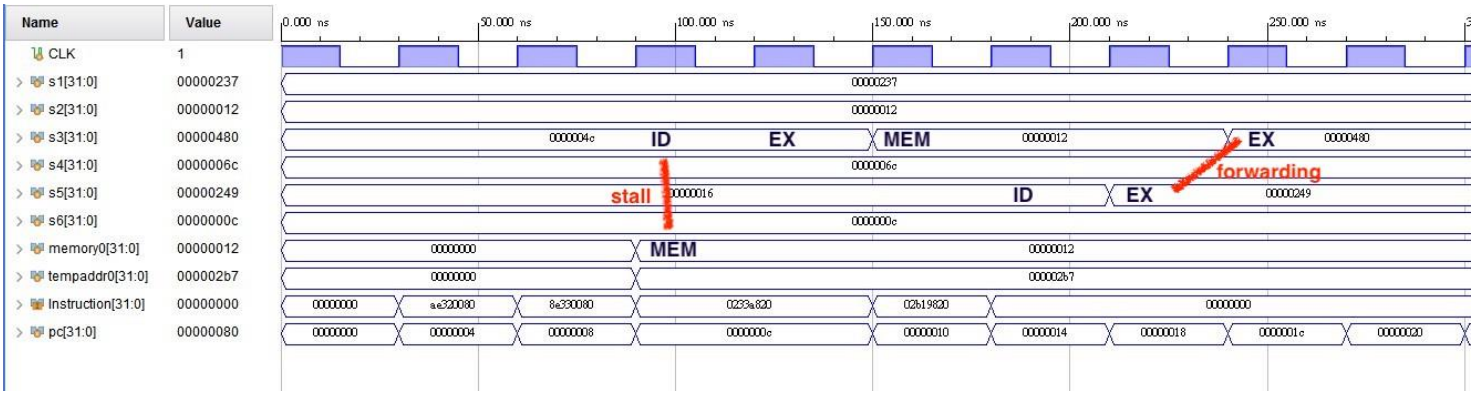

Hazard detection
IF/ID
ID/EX
EX/MEM
MEM/WB
Instruction Fetch
Instruction Decode
Execution
Memory
Write Back
Logic
forwarding

Hazard detection unit use ID/EX.MemRead, IF/ID. RegisisterRt, IF/ID.RegisisterRs and ID/EX.RegisisterRd to determine the program has load use hazard or not.If there are hazard in the program let PC and Instruction memory stay one clock cycle, and let control signal become zero for one cycle.

3. Instruction Set

pc	instruction	machine code
0	NOP	000000_00000_00000_00000_00000_000000
4	sw $2 128($1)	101011_10001_10010_00000_00010_000000
8	lw $3 128($1)	100011_10001_10011_00000_00000_110000
12	add $5 $1 $3	000000_10001_10011_10101_00000_100000
16	add $3 $5 $1	000000_10101_10001_10011_00000_100000
20	NOP	000000_00000_00000_00000_00000_000000

4. 波形圖



References:

Single cycle

GitHub - hxing9974/Verilog-Single-Cycle-Processor: Verilog

GitHub - ybch14/Single-Cycle-CPU-with-Verilog

GitHub - EternalFeather/Single-Cycle-MIPS-CPU: An Implementation of Single cycle MIPS CPU for Computer Orgnization.

Single-cycle Implementation · 課程筆記 (gitbooks.io)

Verilog code for 16-bit single cycle MIPS processor - FPGA4student.com

Pipeline:

GitHub - mhyousefi/MIPS-pipeline-processor: A pipelined implementation of the MIPS processor featuring hazard detection as well as forwarding

GitHub - hxing9974/Verilog-Pipeline-Processor: Verilog

Pipeline CPU Test Bench - Community Forums (xilinx.com)

Pipelined MIPS Processor in Verilog (Part-1) - FPGA4student.com

Hazard Detection:

hazard-detection · GitHub Topics · GitHub

姓名	Fira	Leto	Tavi
比例	33%	33%	34%
工作	設計架構 /RTL code Pipeline	設計架構 /RTL code Hazard detection	設計架構 /RTL code Single cycle 問題分析

註: 書面報告為 3 人合力共同完成，謝謝助教，辛苦了!!